

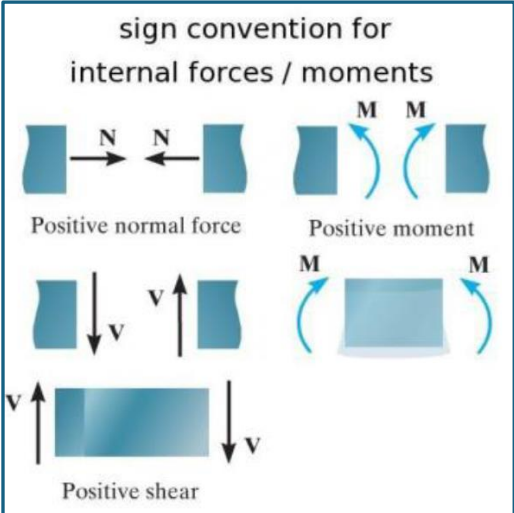
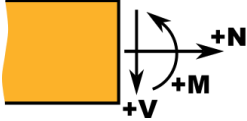
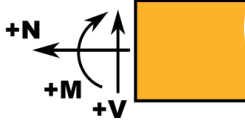
Civ Eng 2200 Statics
Exam 3 Review Sheet

Four Sections:

- 1. Internal Forces
- 2. V & M Diagrams
- 3. Friction (and Wedges and Belts)
- 4. Centroids (Integration, Composite, P&G)

Crib Sheet Formulas per Section:

1. Internal Forces

Crib Sheet:	As we would draw in class:	
sign convention for internal forces / moments  Positive normal force Positive moment Positive shear	From left:	From right:
		

2. V & M Diagrams

When drawing V and M diagrams

 downward jump in M diag.

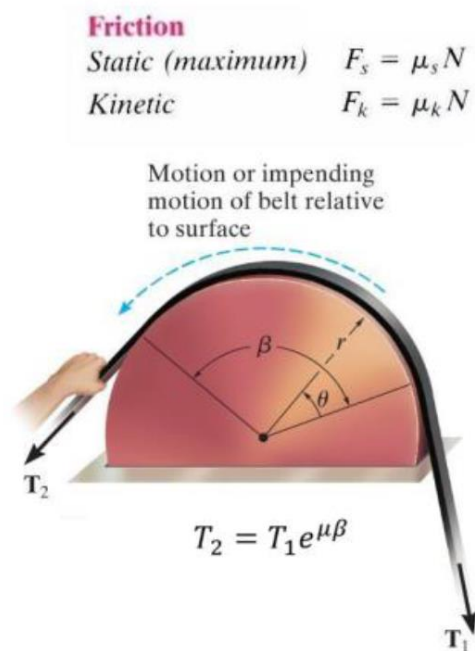
 upward jump in M diag.

NOT on crib sheet but important, you need to memorize: Slope and Area Relations

V&M Slope and Area Relations:

Equation	Explanation
$\frac{dV}{dx} = -w$	At a point, the slope of the V curve = -w.
$\Delta V = - \int w dx$	Over an interval, the change in shear (V) = -Area under Load (w) Curve
$\frac{dM}{dx} = V$	At a point, the slope of the M curve = V
$\Delta M = \int V dx$	Over an interval, the change in Moment (M) = Area under V (Shear) Curve

3. Friction (and Wedges and Belts)

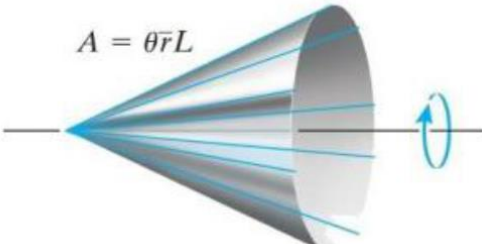
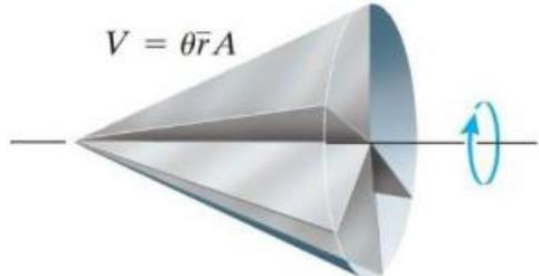


4. Centroids (Integration, Composite, P&G)

<u>Geometric Centroids Volume</u> $\bar{x} = \frac{\int_V \tilde{x} dV}{\int_V dV} \quad \bar{y} = \frac{\int_V \tilde{y} dV}{\int_V dV}$ $\bar{z} = \frac{\int_V \tilde{z} dV}{\int_V dV}$		<u>Geometric Centroids Area</u> $\bar{x} = \frac{\int_A \tilde{x} dA}{\int_A dA} \quad \bar{y} = \frac{\int_A \tilde{y} dA}{\int_A dA}$	
		<u>Geometric Centroids Line Segment</u> $\bar{x} = \frac{\int_L \tilde{x} dL}{\int_L dL} \quad \bar{y} = \frac{\int_L \tilde{y} dL}{\int_L dL}$	

Missing from Crib Sheet: Discrete Formulas, Center of Gravity

Area:	Line:	Volume:	Mass (Center of Gravity)
$\bar{x} = \frac{\sum \tilde{x}_i A_i}{\sum A_i}$	$\bar{x} = \frac{\sum \tilde{x}_i L_i}{\sum L_i}$	$\bar{x} = \frac{\sum \tilde{x}_i V_i}{\sum V_i}$	$\bar{x} = \frac{\sum \tilde{x}_i m_i}{\sum m_i}$

<u>Pappus & Guldinus</u> $A = \theta \bar{r} L$ 	$V = \theta \bar{r} A$ 
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